

HOMEWORK 1
ELEMENTARY LINEAR ALGEBRA
(MATH 2250-03), FALL, 2018

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SECTION 1.1 - SYSTEMS OF LINEAR EQUATIONS

1. Find the augmented matrix associated to the following system of linear equations. Then, using elementary row operations, solve this system.

$$2x_1 + 4x_2 = -4$$

$$5x_1 + 7x_2 = 11$$

The augmented matrix associated to the system of linear equations given in the problem is

$$\left[\begin{array}{cc|c} 2 & 4 & -4 \\ 5 & 7 & 11 \end{array} \right].$$

Observe that elementary row operations allow us to carry out the following reduction:

$$\begin{aligned} \left[\begin{array}{cc|c} 2 & 4 & -4 \\ 5 & 7 & 11 \end{array} \right] &\Rightarrow \left[\begin{array}{cc|c} 2 & 4 & -4 \\ 0 & -6 & 42 \end{array} \right] \text{ (Replace } R_2 \text{ with } 2R_2 - 5R_1) \\ &\Rightarrow \left[\begin{array}{cc|c} 2 & 4 & -4 \\ 0 & 1 & -7 \end{array} \right] \text{ (Replace } R_2 \text{ with } \frac{-1}{6}R_2) \\ &\Rightarrow \left[\begin{array}{cc|c} 1 & 0 & 12 \\ 0 & 1 & -7 \end{array} \right] \text{ (Replace } R_1 \text{ with } \frac{1}{2}R_1 - 2R_2) \end{aligned}$$

Hence, we may conclude that the solution to the system corresponding to the augmented matrix in the last line of the above calculations is $x_1 = 12$, $x_2 = -7$. Moreover, as systems with row-equivalent augmented matrices are equivalent, this solution set coincides with the solution set of our original system.

2. Determine if the following system of linear equations is consistent by finding the augmented matrix associated to the system and solving the system using elementary row operations.

$$\begin{aligned}x_1 + 3x_3 &= 2 \\2x_2 - 3x_4 &= 3 \\-2x_2 + 3x_3 + 2x_4 &= 1 \\3x_1 + 7x_4 &= -5\end{aligned}$$

The augmented matrix associated to the system of linear equations given in the problem is

$$\left[\begin{array}{cccc|c} 1 & 0 & 3 & 0 & 2 \\ 0 & 2 & 0 & -3 & 3 \\ 0 & -2 & 3 & 2 & 1 \\ 3 & 0 & 0 & 7 & -5 \end{array} \right].$$

Observe that elementary row operations allow us to carry out the following reduction:

$$\begin{aligned}\left[\begin{array}{cccc|c} 1 & 0 & 3 & 0 & 2 \\ 0 & 2 & 0 & -3 & 3 \\ 0 & -2 & 3 & 2 & 1 \\ 3 & 0 & 0 & 7 & -5 \end{array} \right] &\Rightarrow \left[\begin{array}{cccc|c} 1 & 0 & 3 & 0 & 2 \\ 0 & 2 & 0 & -3 & 3 \\ 0 & -2 & 3 & 2 & 1 \\ 0 & 0 & -9 & 7 & -11 \end{array} \right] & \text{(Replace } R_4 \text{ with } R_4 - 3R_1) \\ &\Rightarrow \left[\begin{array}{cccc|c} 1 & 0 & 3 & 0 & 2 \\ 0 & 2 & 0 & -3 & 3 \\ 0 & 0 & 3 & -1 & 4 \\ 0 & 0 & -9 & 7 & -11 \end{array} \right] & \text{(Replace } R_3 \text{ with } R_3 + R_2) \\ &\Rightarrow \left[\begin{array}{cccc|c} 1 & 0 & 3 & 0 & 2 \\ 0 & 2 & 0 & -3 & 3 \\ 0 & 0 & 3 & -1 & 4 \\ 0 & 0 & 0 & 4 & 1 \end{array} \right] & \text{(Replace } R_4 \text{ with } R_4 + 3R_3) \\ &\Rightarrow \left[\begin{array}{cccc|c} 1 & 0 & 3 & 0 & 2 \\ 0 & 2 & 0 & -3 & 3 \\ 0 & 0 & 12 & 0 & 17 \\ 0 & 0 & 0 & 4 & 1 \end{array} \right] & \text{(Replace } R_3 \text{ with } 4R_3 + R_4) \\ &\Rightarrow \left[\begin{array}{cccc|c} 1 & 0 & 3 & 0 & 2 \\ 0 & 8 & 0 & 0 & 15 \\ 0 & 0 & 12 & 0 & 17 \\ 0 & 0 & 0 & 4 & 1 \end{array} \right] & \text{(Replace } R_2 \text{ with } 4R_2 + 3R_4) \\ &\Rightarrow \left[\begin{array}{cccc|c} 4 & 0 & 0 & 0 & -9 \\ 0 & 8 & 0 & 0 & 15 \\ 0 & 0 & 12 & 0 & 17 \\ 0 & 0 & 0 & 4 & 1 \end{array} \right] & \text{(Replace } R_1 \text{ with } 4R_1 - R_3) \\ &\Rightarrow \left[\begin{array}{cccc|c} 1 & 0 & 0 & 0 & -9/4 \\ 0 & 1 & 0 & 0 & 15/8 \\ 0 & 0 & 1 & 0 & 17/12 \\ 0 & 0 & 0 & 1 & 1/4 \end{array} \right] & \text{(Divide } R_i \text{ by the leftmost nonzero term in } R_i)\end{aligned}$$

Hence, we may conclude that the solution to the system corresponding to the augmented matrix in the last line of the above calculations is

$$x_1 = -\frac{9}{4}, \quad x_2 = \frac{15}{8}, \quad x_3 = \frac{17}{12}, \quad x_4 = \frac{1}{4}.$$

Moreover, as systems with row-equivalent augmented matrices are equivalent, this solution set coincides with the solution set of our original system. As the original system has a solution (indeed, a unique one), we may conclude that original system is consistent.

3. Do the three planes given by the linear equations $x_1 + 2x_2 + x_3 = 4$, $x_2 - x_3 = 1$ and $x_1 + 3x_2 = 0$ have at least one common point of intersection?

Finding a common point of intersection for the three planes given in the statement of the problem is equivalent to finding a solution to the following linear system of equations:

$$x_1 + 2x_2 + x_3 = 4$$

$$x_2 - x_3 = 1$$

$$x_1 + 3x_2 = 0$$

That is, the linear equations characterizing the three planes in the problem have a common solution precisely when the planes intersect, so the existence of such a point of intersection is equivalent to the consistency of the above system of linear equations.

Note that the augmented matrix associated to the system of linear equations given in the problem is

$$\left[\begin{array}{cccc} 1 & 2 & 1 & 4 \\ 0 & 1 & -1 & 1 \\ 1 & 3 & 0 & 0 \end{array} \right].$$

So, observe that elementary row operations allow us to carry out the following reduction:

$$\begin{aligned} \left[\begin{array}{cccc} 1 & 2 & 1 & 4 \\ 0 & 1 & -1 & 1 \\ 1 & 3 & 0 & 0 \end{array} \right] &\Rightarrow \left[\begin{array}{cccc} 1 & 2 & 1 & 4 \\ 0 & 1 & -1 & 1 \\ 0 & 1 & -1 & -4 \end{array} \right] & \text{(Replace } R_3 \text{ with } R_3 - R_1) \\ &\Rightarrow \left[\begin{array}{cccc} 1 & 2 & 1 & 4 \\ 0 & 1 & -1 & 1 \\ 0 & 0 & 0 & -5 \end{array} \right] & \text{(Replace } R_3 \text{ with } R_3 - R_2) \end{aligned}$$

As the last line of the augmented matrix above corresponds to the inconsistent equation $0 = -5$, the system of linear equations corresponding to the augmented matrix in the last line of the above calculations has no solutions and is correspondingly inconsistent. Hence, as systems with row-equivalent augmented matrices are equivalent, we may conclude that the linear system formed from the equations characterizing the three planes is inconsistent. Therefore, the three planes do not have a common point of intersection.

SECTION 1.2 - ROW REDUCTION AND ECHELON FORMS

4. Using elementary row operations, compute the reduced row echelon form of the following matrix:

$$\begin{bmatrix} 1 & 2 & 3 & 4 \\ 4 & 5 & 6 & 7 \\ 6 & 7 & 8 & 9 \end{bmatrix}$$

Elementary row operations allow us to carry out the following reduction:

$$\begin{aligned} \begin{bmatrix} 1 & 2 & 3 & 4 \\ 4 & 5 & 6 & 7 \\ 6 & 7 & 8 & 9 \end{bmatrix} &\Rightarrow \begin{bmatrix} 1 & 2 & 3 & 4 \\ 0 & 1 & 2 & 3 \\ 6 & 7 & 8 & 9 \end{bmatrix} \quad \left(\text{Replace } R_2 \text{ with } -\frac{1}{3}R_2 + \frac{4}{3}R_1 \right) \\ &\Rightarrow \begin{bmatrix} 1 & 2 & 3 & 4 \\ 0 & 1 & 2 & 3 \\ 0 & 1 & 2 & 3 \end{bmatrix} \quad \left(\text{Replace } R_3 \text{ with } -\frac{1}{5}R_3 + \frac{6}{5}R_1 \right) \\ &\Rightarrow \begin{bmatrix} 1 & 2 & 3 & 4 \\ 0 & 1 & 2 & 3 \\ 0 & 0 & 0 & 0 \end{bmatrix} \quad (\text{Replace } R_3 \text{ with } R_3 - R_2) \\ &\Rightarrow \begin{bmatrix} 1 & 0 & -1 & -2 \\ 0 & 1 & 2 & 3 \\ 0 & 0 & 0 & 0 \end{bmatrix} \quad (\text{Replace } R_1 \text{ with } R_1 - 2R_2) \end{aligned}$$

Thus, the reduced row echelon form of the matrix given in the statement of the problem is

$$\begin{bmatrix} 1 & 0 & -1 & -2 \\ 0 & 1 & 2 & 3 \\ 0 & 0 & 0 & 0 \end{bmatrix}.$$

5. Find the general solution of the system of linear equations that corresponds to the following augmented matrix:

$$\begin{bmatrix} 1 & -3 & 0 & -1 & 0 & -2 \\ 0 & 1 & 0 & 0 & -4 & 1 \\ 0 & 0 & 0 & 1 & 9 & 4 \\ 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$$

The system associated to the augmented matrix given in the statement of the problem is

$$x_1 - 3x_2 - x_4 = -2$$

$$x_2 - 4x_5 = 1$$

$$x_4 + 9x_5 = 4$$

$$0 = 0$$

Elementary row operations allow us to carry out the following reduction on the augmented matrix:

$$\begin{bmatrix} 1 & -3 & 0 & -1 & 0 & -2 \\ 0 & 1 & 0 & 0 & -4 & 1 \\ 0 & 0 & 0 & 1 & 9 & 4 \\ 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix} \Rightarrow \begin{bmatrix} 1 & 0 & 0 & -1 & -12 & 1 \\ 0 & 1 & 0 & 0 & -4 & 1 \\ 0 & 0 & 0 & 1 & 9 & 4 \\ 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix} \quad (\text{Replace } R_1 \text{ with } R_1 + 3R_2)$$

$$\Rightarrow \begin{bmatrix} 1 & 0 & 0 & 0 & -3 & 5 \\ 0 & 1 & 0 & 0 & -4 & 1 \\ 0 & 0 & 0 & 1 & 9 & 4 \\ 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix} \quad (\text{Replace } R_1 \text{ with } R_1 + R_3)$$

One may note that the final augmented matrix given above is the reduced row-echelon form of the original augmented matrix. This corresponds to the system

$$x_1 - 3x_5 = 5$$

$$x_2 - 4x_5 = 1$$

$$x_4 + 9x_5 = 4$$

$$0 = 0$$

Further, we may note that x_3 and x_5 are free variables, as the third and fifth column of the reduced row echelon form of the augmented matrix do not contain a pivot. As such, this then allows us to give a parametric description of the solution set as follows:

$$x_3, x_5 \in \mathbb{R}, \quad x_4 = 4 - 9x_5, \quad x_2 = 1 + 4x_5, \quad x_1 = 5 + 3x_5.$$

More compactly, we could also characterize the solution set as follows:

$$\left\{ \begin{bmatrix} 5 + 3x_5 \\ 1 + 4x_5 \\ x_3 \\ 4 - 9x_5 \\ x_5 \end{bmatrix} \in \mathbb{R}^5 : x_3, x_5 \in \mathbb{R} \right\}$$

6. Determine the value(s) of $h \in \mathbb{R}$ such that the following matrix is the augmented matrix of a consistent linear system:

$$\begin{bmatrix} 2 & 3 & h \\ 4 & 6 & 7 \end{bmatrix}$$

Elementary row operations allow us to carry out the following reduction on the augmented matrix:

$$\begin{bmatrix} 2 & 3 & h \\ 4 & 6 & 7 \end{bmatrix} \Rightarrow \begin{bmatrix} 2 & 3 & h \\ 0 & 0 & 7 - 2h \end{bmatrix} \quad (\text{Replace } R_2 \text{ with } R_2 - 2R_1)$$

Note that the system corresponding to the augmented matrix in the last line of the above calculations is

$$\begin{aligned} 2x_1 + 3x_2 &= h \\ 0 &= 7 - 2h \end{aligned}$$

Clearly, this system is inconsistent unless $7 - 2h = 0$. Moreover, as x_2 is a free variable, the first equation in this system has a solution for any value of h . Thus, we may conclude that the only value of h for which the augmented matrix given in the statement of the problem corresponds to a consistent linear system is $h = \frac{7}{2}$.

SECTION 1.3 - VECTOR EQUATIONS

7. Let $a_1 = \begin{bmatrix} 1 \\ 4 \\ -2 \end{bmatrix}$, $a_2 = \begin{bmatrix} -2 \\ -3 \\ 7 \end{bmatrix}$, and $b = \begin{bmatrix} 4 \\ 1 \\ h \end{bmatrix}$. For what value(s) of $h \in \mathbb{R}$ is $b \in \text{span}\{a_1, a_2\}$?

If $b \in \text{span}\{a_1, a_2\}$, then $c_1 a_1 + c_2 a_2 = b$ for some constants $c_1, c_2 \in \mathbb{R}$. That is, $b \in \text{span}\{a_1, a_2\}$ if

$$\begin{aligned} c_1 a_1 + c_2 a_2 &= b \\ c_1 \begin{bmatrix} 1 \\ 4 \\ -2 \end{bmatrix} + c_2 \begin{bmatrix} -2 \\ -3 \\ 7 \end{bmatrix} &= \begin{bmatrix} 4 \\ 1 \\ h \end{bmatrix} \\ \begin{bmatrix} c_1 - 2c_2 \\ 4c_1 - 3c_2 \\ -2c_1 + 7c_2 \end{bmatrix} &= \begin{bmatrix} 4 \\ 1 \\ h \end{bmatrix} \end{aligned}$$

So, $b \in \text{span}\{a_1, a_2\}$ is equivalent to the consistency of the system of linear equations

$$\begin{aligned} c_1 - 2c_2 &= 4 \\ 4c_1 - 3c_2 &= 1 \\ -2c_1 + 7c_2 &= h \end{aligned}$$

Note that the augmented matrix associated to this system of linear equations is

$$\left[\begin{array}{cc|c} 1 & -2 & 4 \\ 4 & -3 & 1 \\ -2 & 7 & h \end{array} \right].$$

So, observe that elementary row operations allow us to carry out the following reduction:

$$\begin{aligned} \left[\begin{array}{cc|c} 1 & -2 & 4 \\ 4 & -3 & 1 \\ -2 & 7 & h \end{array} \right] &\Rightarrow \left[\begin{array}{cc|c} 1 & -2 & 4 \\ 4 & -3 & 1 \\ 0 & 11 & 2h+1 \end{array} \right] \quad (\text{Replace } R_3 \text{ with } 2R_3 + R_2) \\ &\Rightarrow \left[\begin{array}{cc|c} 1 & -2 & 4 \\ 0 & 1 & -3 \\ 0 & 11 & 2h+1 \end{array} \right] \quad \left(\text{Replace } R_2 \text{ with } \frac{1}{5}R_2 - \frac{4}{5}R_1 \right) \\ &\Rightarrow \left[\begin{array}{cc|c} 1 & 0 & -2 \\ 0 & 1 & -3 \\ 0 & 11 & 2h+1 \end{array} \right] \quad (\text{Replace } R_1 \text{ with } R_1 + 2R_2) \\ &\Rightarrow \left[\begin{array}{cc|c} 1 & 0 & -2 \\ 0 & 1 & -3 \\ 0 & 0 & 2h+34 \end{array} \right] \quad (\text{Replace } R_3 \text{ with } R_3 - 11R_2) \end{aligned}$$

Clearly, this system is inconsistent unless $2h + 34 = 0$. Moreover, as the first two equations in the system corresponding to the augmented matrix in the final line of the calculation given above have a solution for any value of h . So, the linear system corresponding to our original augmented matrix is consistent when $h = -17$. As such, we may conclude that $b \in \text{span}\{a_1, a_2\}$ if and only if $h = -17$.

8. Let $v_1, \dots, v_k \in \mathbb{R}^3$ and suppose that for $j = 1, \dots, k$ an object with mass $m_j \in \mathbb{R}$ is located at v_j . Physicists call such objects *point masses*. The total mass of this system of point masses is $m = m_1 + \dots + m_k$, and the center of mass of the system is $\bar{v} = \frac{1}{m}(m_1 v_1 + \dots + m_k v_k)$. Find the center of mass for the following system of point masses:

$$v_1 = \begin{bmatrix} 5 \\ -4 \\ 3 \end{bmatrix}, \quad v_2 = \begin{bmatrix} 4 \\ 3 \\ -2 \end{bmatrix}, \quad v_3 = \begin{bmatrix} -4 \\ -3 \\ -1 \end{bmatrix}, \quad v_4 = \begin{bmatrix} -9 \\ -8 \\ 6 \end{bmatrix}$$

$$m_1 = 2g, \quad m_2 = 5g, \quad m_3 = 2g, \quad m_4 = 1g$$

To find the center of mass for the system of point masses given in the statement of the problem, we first calculate the total mass of the system:

$$\begin{aligned} m &= m_1 + m_2 + m_3 + m_4 \\ &= 2g + 5g + 2g + 1g \\ &= 10g. \end{aligned}$$

Following this, we may now directly compute the center of mass:

$$\begin{aligned} \bar{v} &= \frac{1}{m}(m_1 v_1 + m_2 v_2 + m_3 v_3 + m_4 v_4) \\ &= \frac{1}{10g} \left(2g \begin{bmatrix} 5 \\ -4 \\ 3 \end{bmatrix} + 5g \begin{bmatrix} 4 \\ 3 \\ -2 \end{bmatrix} + 2g \begin{bmatrix} -4 \\ -3 \\ -1 \end{bmatrix} + 1g \begin{bmatrix} -9 \\ -8 \\ 6 \end{bmatrix} \right) \\ &= \frac{2g}{10g} \begin{bmatrix} 5 \\ -4 \\ 3 \end{bmatrix} + \frac{5g}{10g} \begin{bmatrix} 4 \\ 3 \\ -2 \end{bmatrix} + \frac{2g}{10g} \begin{bmatrix} -4 \\ -3 \\ -1 \end{bmatrix} + \frac{1g}{10g} \begin{bmatrix} -9 \\ -8 \\ 6 \end{bmatrix} \\ &= \frac{1}{5} \begin{bmatrix} 5 \\ -4 \\ 3 \end{bmatrix} + \frac{1}{2} \begin{bmatrix} 4 \\ 3 \\ -2 \end{bmatrix} + \frac{1}{5} \begin{bmatrix} -4 \\ -3 \\ -1 \end{bmatrix} + \frac{1}{10} \begin{bmatrix} -9 \\ -8 \\ 6 \end{bmatrix} \\ &= \begin{bmatrix} 1 \\ -4/5 \\ 3/5 \end{bmatrix} + \begin{bmatrix} 2 \\ 3/2 \\ -1 \end{bmatrix} + \begin{bmatrix} -4/5 \\ -3/5 \\ -1/5 \end{bmatrix} + \begin{bmatrix} -9/10 \\ -4/5 \\ 3/5 \end{bmatrix} \\ &= \begin{bmatrix} 13/10 \\ -7/10 \\ 0 \end{bmatrix}. \end{aligned}$$